petitpy

Python where CPython doesn't fit.

A small Python runtime, written in Zig. Under a megabyte on your machine, about 350 KB in a browser tab.

~/demo

```
~/demo ❯ cat where.py
from dataclasses import dataclass

@dataclass
class Point:
    x: int
    y: int

def where(p):
    match p:
        case Point(x=0, y=0):
            return "origin"
        case Point(x=0, y=y):
            return f"on the y axis, at {y}"
        case _:
            return f"elsewhere: {p}"

for p in [Point(0, 0), Point(0, 5), Point(3, 4)]:
    print(where(p))

~/demo ❯ ppy where.py
origin
on the y axis, at 5
elsewhere: Point(x=3, y=4)
~/demo ❯
```

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TODAY

Python is either big or partial

In a browser tab

Pyodide brings all of CPython to the page: tens of megabytes before the first `print`.

On a small server

A stdlib-only daemon sits at 17 MB of RAM on CPython, on a box with little to spare.

On MicroPython

Small and quick, but made for microcontrollers. Code written for CPython trips on what it leaves out.

petitpy keeps most of CPython's language at MicroPython's size.

IN THE BROWSER

Python in a browser tab

About 350 KB

170 KB gzipped, so a page loads it in well under a second.

The page, from Python

`import js` reaches the DOM, and Python functions answer the page's events.

Games and demos

Breakout, Tetris, 2048 and more, written in Python, at petitpy.lab.abilian.com/play.



ONE RUNTIME

The same Python, in the terminal and in the page

Write it

`app.py`

Python 3, with `match`, f-strings, dataclasses and exception chaining

→

Run it

`ppy app.py`

on macOS or Linux, arm64 or x86_64, from one static binary

→

Embed it

`py.run(source)`

in a web page or in Node, from the npm package

It shares MicroPython's bytecode too: an `.mpy` file compiled by one runs on the other.

MEASURED

Small, light, quick and tested

Small

A static binary under 1 MB, with a standard library of about 200 KB on disk.

Quick

Faster than MicroPython on our benchmarks, at about 1.4 times CPython's time.

Light

A real system daemon runs in 4 MB of RAM, where CPython needs 17 MB.

Tested

MicroPython's own test suite passes byte for byte on every commit, next to tests compared with CPython.

ALPHA, AVAILABLE NOW

Try it today

macOS · Linux · your browser

```
uvx petitpy script.py
pip install petitpy
npm install petitpy
```

The first line runs a script without installing anything.

Docs: petitpy.lab.abilian.com

Source: github.com/abilian/petitpy

Play: petitpy.lab.abilian.com/play

It's an alpha. The docs list where petitpy differs from CPython on purpose; anything else is worth a report.

Thanks to MicroPython, by Damien P. George and contributors, where petitpy started. MIT license. Made by Abilian in France.